

Flight-Test Characterization and Design Implications of Quadplane Hover-to-Forward-Flight Transition

Md Azizul Islam*

KaiLab LLC, Sugar Land, TX 77479, United States

Md Nur-A-Adam Dony†

Tennessee Technological University, Cookeville, TN 38505, United States

Mehedi Hasan‡

University of Arkansas, Fayetteville, AR 72701, United States

S. G. M. Hossain§

Blue Origin, Kent, WA 98032, United States

Hybrid vertical take-off and landing aircraft that combine multicopter lift with a fixed wing pair the operational flexibility of a multicopter with the cruise efficiency of an airplane, at the cost of a mode-switching transition that must remain reliably stable. This paper reports a flight-test characterization of the hover phase, the hover-to-forward-flight transition, and the fixed-wing cruise of a small quadplane, using telemetry recorded by an open-source autopilot, complemented by a computational aerodynamic study of the wing. Body angular rates, attitude excursions, ground speed, electrical power, bank-angle distributions, and elevator trim are extracted from the on-board estimator and cross-validated against numerical derivatives of the recorded attitude quaternion. The aircraft exhibits a stationary hover, a smooth front transition completing at an airspeed of 11.9 meters per second, and a back transition completing in 1.86 seconds, with peak body-axis rates well within control authority. The hover-to-cruise electrical-power ratio is 3.36. An angle-of-attack sweep of a finite wing panel resolves the lift and drag polars, identifies the onset of stall, and quantifies an approximately 33 percent lift loss under a quasi-steady airspeed deficit, anchoring an operational wind ceiling. Design rules for airspeed margin, control authority, power sizing, energy budget, and wind tolerance are derived.

Nomenclature

*Research and Development Specialist, UAV Design and Manufacture; azizul@kailab.com.

†Ph.D. Candidate, Department of Electrical and Computer Engineering; mdony42@tntech.edu (Corresponding Author).

‡Graduate Research Assistant, Department of Mechanical Engineering; mehedi@uark.edu.

§Senior Manufacturing Engineer, Lunar Permanence Program; smgmamur@yahoo.com.

ϕ, θ, ψ	=	roll, pitch, yaw angles, rad (deg where stated)
p, q, r	=	body-axis angular rates, rad/s (deg/s where stated)
q_w, q_x, q_y, q_z	=	unit attitude quaternion components
V_N, V_E, V_D	=	north-east-down (NED) frame ground velocity, m/s
V_h	=	horizontal ground speed, m/s
V_{air}	=	indicated airspeed (IAS) from pitot, m/s
$V_T, V_{T,\text{back}}$	=	front and back transition airspeeds, m/s
V_c	=	cruise airspeed, m/s
$h, \dot{h}$	=	altitude above ground level and climb rate, m, m/s
$\delta_e, \delta_a, \delta_r$	=	elevator, aileron, rudder deflection, deg
R	=	turn radius, m
P, E	=	electrical power and energy, W, W·h
V_{bat}, I	=	battery pack voltage and current, V, A
L, D, T	=	aerodynamic lift, drag, and thrust, N
C_L, C_D, C_M	=	lift, drag, and pitching-moment coefficients
C_{L_α}	=	lift-curve slope, rad^{-1}
α	=	angle of attack, rad (deg where stated)
n	=	load factor
W	=	aircraft weight, N
g	=	gravitational acceleration, m/s^2
ρ, μ	=	air density and dynamic viscosity, kg/m^3 , $\text{Pa}\cdot\text{s}$
S, c, b	=	wing reference area, chord, and span, m^2 , m, m
Re	=	chord Reynolds number
$\eta_{\text{prop}}, \eta_{\text{ESC}}$	=	propeller and electronic-speed-controller efficiency
N_r	=	number of lift rotors
A_d	=	single-rotor disc area, m^2
σ	=	sample standard deviation

I. Introduction

A. Background and Motivation

HYBRID VTOL UAVs combine the runway-independent operation of a multicopter with the cruise efficiency of a fixed-wing aircraft, and have become a preferred platform for medium-range surveillance, mapping, and logistics

missions [1, 2]. The development of small UAV airframes has progressed from early FPGA-based remote-piloted platforms [3] and digital multi-rotor control systems [4] toward autonomous open-source autopilot ecosystems, and wing design choices for small UAVs now span fixed cambered low-Reynolds-number sections to more sophisticated morphing-airfoil concepts [5–7]. Among hybrid configurations, the *quadplane*—a conventional airplane retrofitted with four dedicated lift rotors—offers the simplest control architecture: the multicopter (MC) and fixed-wing (FW) flight modes are mechanically decoupled, and the autopilot manages a single coordinated handover during transition. The trade-off is that the lift rotors carry parasitic mass and drag in cruise, while the wing and pusher are unused in hover, so the platform must be efficient enough in cruise to amortize the parasitic systems [8]. The test vehicle used in this study is shown at rest in Fig. 1, with the composite-skinned airframe, twin lift booms, and rear pusher motor visible.



Fig. 1 Test vehicle at rest: a small quadplane with four boom-mounted lift rotors and a rear pusher propeller.

The most demanding flight regime is the *front transition* (MC→FW), during which the wing must develop sufficient lift to support the aircraft before the lift rotors are commanded off. If transition is initiated too slowly, the wing stalls; if too quickly, the lift rotors saturate while fighting decelerating wing downwash [9, 10]. Practical PX4-class autopilots manage this trade through a single threshold airspeed V_T at which the controller commits to FW.* Robust transition is therefore a stability problem first—the airframe must remain trim during a regime change in which control authority transfers from rotors to aerodynamic surfaces—and a design problem second—the airframe must possess sufficient power, control, and aerodynamic margins to accommodate the worst-case excursion.

*PX4 User Guide: VTOL Aircraft Configuration, https://docs.px4.io/main/en/frames_vtol/ [retrieved 12 May 2026].

B. Related Work

The literature on small VTOL UAV transition control spans three configurations. **Tail-sitter aircraft** have dominated empirical work: Beach et al. [11] introduced a resolved tilt-twist attitude representation that avoids the Euler-angle singularities of a 90° pitch swing, Lyu et al. [12] demonstrated hierarchical attitude control on a quadrotor tail-sitter, Verling et al. [13] reported full-attitude transition on a flying-wing tail-sitter, Kubo and Suzuki [14] characterized tail-sitter transitional flight, and Ritz and D’Andrea [15] introduced a global flight-envelope controller for flying-wing tail-sitters. **Tilt-rotor and tilt-wing aircraft** have been studied by Maqsood and Go [9] on transition optimization, Geuther et al. [16] on a NASA tandem tilt-wing demonstrator, Yeh et al. [17] on an inverse machine-learning approach to tilt-wing takeoff, and May et al. [18] on tilt-wing transition strategies. **Quadplanes** have received comparatively less empirical attention: Mathur and Atkins [19] reported an experimental aerodynamic analysis of a quadplane UAS in the wind tunnel, Lee and Kim [20] proposed MPC-based longitudinal control for a fixed-wing VTOL, Liu et al. [10] demonstrated INDI transition control on a small VTOL, and Chiwande et al. [21] presented adaptive transition control for quadplane VTOLs in urban gust environments. Survey work [1, 2, 8] consistently identifies the lift+cruise quadplane as the simplest and most energy-efficient hybrid topology for medium-range missions. The control-theoretic background for VTOL and related aerospace platforms is a parallel theme: parametric-uncertainty tracking control for VTOL vehicles has been developed in [22], distributed formation control of spacecraft and higher-order-dynamics UAVs in [23, 24], command-filtered robust tracking on marine and surface platforms in [25], and safe reinforcement-learning control with control barrier functions in [26, 27], providing complementary algorithmic context for the experimental characterization reported here.

C. Comparison with Existing Methods and Contributions

Three characteristics of prior empirical literature motivate the present work. *First*, many published flight-test characterizations rely on proprietary autopilot stacks whose raw telemetry is not openly accessible. *Second*, the bulk of VTOL transition work is simulation-led, with flight data appearing as a single validation point rather than the primary characterization tool. *Third*, the configuration-specific literature is heavily weighted toward tail-sitter and tilt-rotor designs, while empirical studies that report stability margins, transition timing, power budgets, and aerodynamic interpretation *jointly* on a quadplane platform are sparse.

Table 2 summarizes the scope of representative prior studies and contrasts each with the present work along eight axes: aircraft configuration, methodology, primary data source, transition-airspeed reporting, transition-duration reporting, power-budget reporting, lateral-directional characterization, and CFD aerodynamic interpretation. No prior study in the table simultaneously satisfies the joint empirical+CFD+design-rule scope on a quadplane platform.

This paper contributes the following, none of which is jointly addressed in the prior literature reviewed:

- 1) **Configuration focus on quadplanes.** Where prior empirical transition work largely targets tail-sitters [11–15]

Table 2 Comparison of the present work with representative prior literature on small VTOL UAV transition. ✓ = reported in primary text; ✗ = not reported. Configuration codes: TS = tail-sitter; FW = flying-wing; Quad. = quadrotor. Data codes: “Emp.” = empirical (real-flight); “Sim.” = simulation; “WT” = wind tunnel.

Reference	Config.	Methodology	Data	V_T	Dur.	Pow.	Lat./dir.	CFD
Beach 2014 [11]	TS	Tilt-twist attitude	Sim.+flight	✗	✗	✗	✗	✗
Lyu 2017 [12]	Quad. TS	Hierarchical attitude	Emp.	✓	✓	✗	partial	✗
Verling 2016 [13]	FW TS	Full-attitude	Emp.	✓	✓	✗	✓	✗
Kubo 2008 [14]	TS	Transitional flight	Emp.+WT	✓	✓	✗	✗	WT
Ritz 2017 [15]	FW TS	Global envelope	Emp.	✓	✓	✗	✓	✗
Maqsood 2012 [9]	Tilt-rotor	Aerodynamic vectoring	Sim.	✓	✓	✗	✗	Invisc.
Geuther 2020 [16]	Tilt-wing	Sub-scale free-flight	Emp.	✓	partial	✗	partial	✗
Yeh 2023 [17]	Tilt-wing	Inverse-ML takeoff	Sim.	✓	✗	✗	✗	✗
May 2024 [18]	Tilt-wing	Transition strategy	Sim.	✓	✓	✗	partial	✗
Mathur 2023 [19]	QuadPlane	Wind-tunnel aerodynamics	WT	✗	✗	✗	✗	✓ (WT)
Lee 2024 [20]	FW VTOL	MPC longitudinal	Sim.	✓	✗	✗	✗	✗
Chiwande 2025 [21]	QuadPlane	INAV adaptive gust	Emp.	partial	✗	✗	partial	✗
This work	Quadplane	PX4 sdlog2 + Fluent CFD	Emp.+CFD	✓	✓	✓	✓	✓ (CFD)

or tilt-wing and tilt-rotor airframes [9, 16–18], this paper provides a flight-test characterization of a small quadplane—a configuration that remains under-represented in the empirical literature [1, 19].

- 2) **Open, reproducible methodology.** The telemetry pipeline operates exclusively on the open-source PX4 sdlog2 record. Every derived state—quaternion-to-Euler, body rates, ground speed, wind, power, energy, bank, turn radius, elevator trim—is cross-validated against firmware-stored fields, and per-topic CSV bundles are released (see Data Availability). The wind-tunnel work of Mathur and Atkins [19] provides aerodynamic data on a comparable platform; the present approach complements it with real-flight evidence.
- 3) **Joint longitudinal and lateral-directional characterization on a single flight.** Body-rate, attitude, transition-timing, bank-angle distribution, coordinated-turn cross-check, and elevator-trim results are derived from the same flight log. No reviewed quadplane study reports all six on real flight data.
- 4) **CFD aerodynamic interpretation with gust anchoring.** An Ansys Fluent angle-of-attack sweep of a finite-span S1223 wing panel gives $C_L(\alpha)$ and $C_D(\alpha)$ polars with a quantitative drag-polar decomposition (zero-lift drag and Oswald factor), multi-AoA pressure contours, flow visualization, and a quasi-steady airspeed-deficit case that converts the empirical V_T threshold into a physically anchored, gust-margin-quantified design quantity.
- 5) **Inequality-form design rules.** Airspeed margin, control authority, power-system sizing, energy budget, and operational wind ceiling are translated from the combined empirical and CFD evidence into compact design inequalities (Sec. VIII).

The results derive from a single flight on one airframe and constitute a case study; broader statistical validation through additional flights and platforms is identified as the central future-work theme (Sec. IX).

The remainder of the paper is organized as follows. Section II presents the mathematical preliminaries. Section III

describes the data-extraction and CFD methodologies, together with the aerodynamic and energetic relations specific to the quadplane configuration. Sections IV and V describe the test vehicle and the experimental setup. Section VI reports the flight-test results. Section VII presents the computational aerodynamic analysis. Section VIII extracts the design implications, and Section IX concludes.

II. Preliminaries

This section establishes the mathematical framework used to derive aircraft states from the recorded PX4 telemetry and from the complementary CFD sweep. All symbols are defined in the Nomenclature.

A. Attitude Representation

The aircraft attitude is stored on board as a unit quaternion $\mathbf{q} = (q_w, q_x, q_y, q_z)$ with $q_w^2 + q_x^2 + q_y^2 + q_z^2 = 1$. The corresponding Tait–Bryan Euler angles (yaw–pitch–roll, Z - Y - X intrinsic rotation) follow [28–30]

$$\phi = \text{atan2}(2(q_w q_x + q_y q_z), 1 - 2(q_x^2 + q_y^2)), \quad (1)$$

$$\theta = \arcsin[2(q_w q_y - q_z q_x)], \quad (2)$$

$$\psi = \text{atan2}(2(q_w q_z + q_x q_y), 1 - 2(q_y^2 + q_z^2)) \quad (3)$$

B. Body Angular-Rate Kinematics

The relationship between body-axis angular rates (p, q, r) and Euler-angle rates is [28, 30]

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\sin \theta \\ 0 & \cos \phi & \sin \phi \cos \theta \\ 0 & -\sin \phi & \cos \phi \cos \theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}, \quad (4)$$

which is valid in the non-singular range $|\theta| < \pi/2$.

C. Ground Speed, Airspeed, and Wind

The horizontal ground speed is $V_h = \sqrt{V_N^2 + V_E^2}$ and the climb rate is $\dot{h} = -V_D$. In a stationary hover ($V_h \approx 0$), the steady wind component along the aircraft body- x axis is

$$V_{\text{wind}} \approx V_{\text{air}} - V_h \quad (5)$$

D. Electrical Power, Energy, and Endurance

Instantaneous electrical power, accumulated energy, and pure-phase endurance are

$$P(t) = V_{\text{bat}}(t) I(t), \quad E = \int_{t_0}^{t_1} P(t) dt, \quad t_{\text{phase}} = \frac{E_{\text{pack}}}{\bar{P}_{\text{phase}}}, \quad (6)$$

and the mission-average power for a hover-duty fraction η_h is

$$\bar{P}_{\text{mission}} = \eta_h \bar{P}_{\text{hover}} + (1 - \eta_h) \bar{P}_{\text{cruise}} \quad (7)$$

E. Coordinated Turn and Load Factor

In a coordinated turn at airspeed V and bank angle ϕ , the horizontal force balance yields [29, 31]

$$R = \frac{V^2}{g \tan \phi}, \quad \dot{\psi} = \frac{g \tan \phi}{V}, \quad n = \frac{1}{\cos \phi}, \quad (8)$$

with n the load factor. The turn radius R can be cross-checked kinematically from the observed yaw rate as $R = V_h/|\dot{\psi}|$; agreement of the two estimates indicates negligible side-slip.

F. Elevator Trim

For a standard servo with neutral pulse-width-modulation (PWM) command 1500 μs and full deflection $\pm\delta_{\text{max}}$ at $\pm 500 \mu\text{s}$, the commanded elevator angle is

$$\delta_e [\text{deg}] = \frac{\text{PWM} - 1500}{500} \delta_{\text{max}} \quad (9)$$

G. Aerodynamic Forces and Reynolds Number

Lift, drag, and pitching moment on a wing of reference area S and chord c at airspeed V are

$$L = \frac{1}{2} \rho V^2 S C_L(\alpha), \quad D = \frac{1}{2} \rho V^2 S C_D(\alpha), \quad M = \frac{1}{2} \rho V^2 S c C_M(\alpha) \quad (10)$$

In the linear regime, $C_L(\alpha) \approx C_{L_0} + C_{L_\alpha} \alpha$ with $C_{L_\alpha} = dC_L/d\alpha$ the lift-curve slope. The drag polar follows the parabolic approximation

$$C_D(\alpha) \approx C_{D_0} + k C_L^2(\alpha), \quad (11)$$

with C_{D_0} the zero-lift drag coefficient and k the induced-drag factor. The chord Reynolds number is

$$Re = \frac{\rho V c}{\mu}, \quad (12)$$

and the lift-to-drag ratio is

$$\frac{L}{D} = \frac{C_L(\alpha)}{C_D(\alpha)}, \quad (13)$$

maximized when $dC_D/dC_L = C_D/C_L$. Low-Reynolds-number effects on $C_{L,\max}$ and drag are reviewed in [32–34].

III. Methodology

This section describes the methodology in four parts: the binary-log data-extraction pipeline (Sec. III.A); the numerical operators that produce the results of Sec. VI (Sec. III.B); the aerodynamic and energetic balance equations that connect the flight measurements to the airframe’s design quantities (Sec. III.C); and the Ansys Fluent setup used for the complementary aerodynamic sweep of Sec. VII (Sec. III.D). The end-to-end methodology is summarized in Fig. 2.

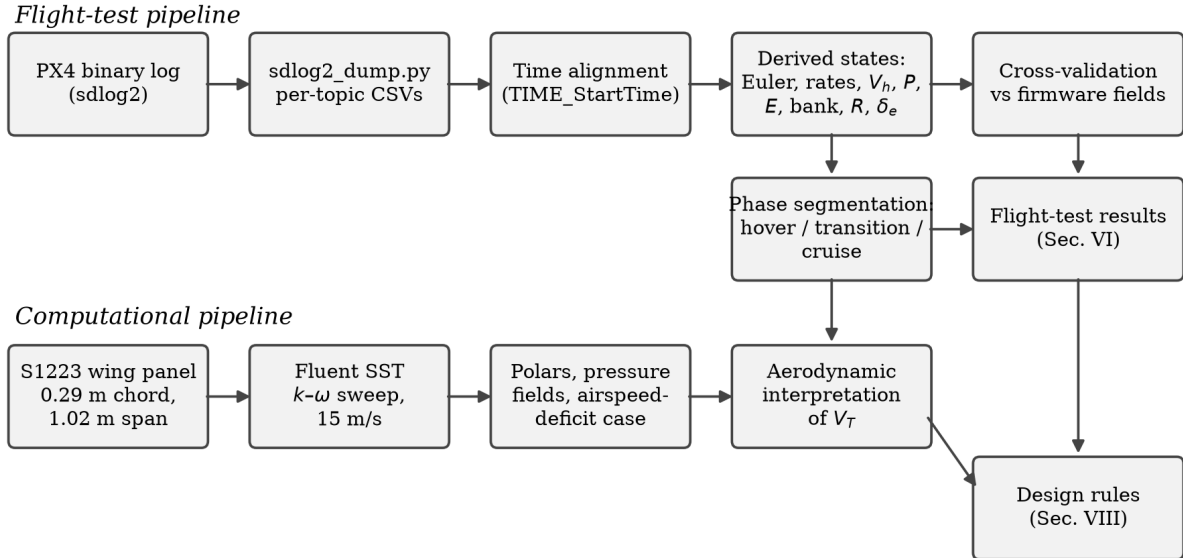


Fig. 2 End-to-end methodology: the flight-test pipeline from binary log to phase-resolved results, and the computational pipeline from wing-panel geometry to the aerodynamic interpretation, converging on the design rules.

A. Data Extraction from the sdlog2 Binary Format

The PX4 binary log used in this study is in the legacy sdlog2 format, identified by the file-leading magic bytes `0xA3 0x95 0x80`. Each message is preceded by a three-byte header (`0xA3 0x95 <MSG_TYPE>`); the FMT message (`<MSG_TYPE>=0x80`) declares the binary layout of all subsequent message types. The file is parsed using

sdlog2_dump.py from the PX4-Autopilot v1.7.0 release tag, which produces per-topic CSV outputs filtered with the -m option. Global time alignment is performed by forward-filling the TIME_StartTime field, giving every row a microsecond-precision wall-time coordinate.

The following topics are used: ATT (attitude quaternion and body rates, ~ 250 Hz); ATSP (attitude setpoint); AIRS (calibrated indicated airspeed, ~ 1 kHz); BATT (filtered pack voltage and current); LPOS and LPSP (local position estimator and setpoint); GPOS (global position); OUT0 and OUT1 (PWM commands); and VTOL (mode flags). Euler angles obtained from Eqs. (1)–(3) are verified against the firmware-stored ATT_Pitch, ATT_Roll, and ATT_Yaw fields to residuals below $2 \times 10^{-6}^\circ$, and body rates from ATT_*Rate are cross-validated against numerical Euler-angle derivatives obtained via Eq. (4).

B. Numerical Operators on Per-Topic Samples

The CSV samples are non-uniform. Numerical derivatives are computed by centred differences,

$$\dot{x}_i \approx \frac{x_{i+1} - x_{i-1}}{t_{i+1} - t_{i-1}} \quad (14)$$

Phase-window integrals [Eq. (6)] are evaluated by the trapezoidal rule,

$$E \approx \sum_{i=0}^{N-1} \frac{1}{2} (P(t_i) + P(t_{i+1})) (t_{i+1} - t_i) \quad (15)$$

Per-phase statistics use the unbiased sample estimators

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i, \quad \sigma_x^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2. \quad (16)$$

A state is reported as “essentially zero in the mean” when $|\bar{x}| < \sigma_x$.

Phase windows are identified using the recorded rotor-wing and transition flags. Let $f_{\text{RW}}(t)$ and $f_{\text{Tr}}(t)$ denote the rotor-wing and transition indicators. The phase logic is

$$\text{phase}(t) = \begin{cases} \text{MC hover} & f_{\text{RW}} = 1, f_{\text{Tr}} = 0, V_h < 0.5 \text{ m/s}, \\ \text{MC} \rightarrow \text{FW transition} & f_{\text{Tr}} = 1, \dot{f}_{\text{RW}} < 0, \\ \text{FW cruise} & f_{\text{RW}} = 0, f_{\text{Tr}} = 0, \\ \text{FW} \rightarrow \text{MC transition} & f_{\text{Tr}} = 1, \dot{f}_{\text{RW}} > 0, \end{cases} \quad (17)$$

with the MC $\rightarrow$ FW and FW $\rightarrow$ MC branches distinguished by the sign of the rotor-wing edge. The cruise window for elevator-trim analysis is further restricted to samples satisfying the level-flight criterion $|\phi| < 5^\circ$ and $|\dot{h}| < 0.5$ m/s.

C. Aerodynamic and Energetic Balance Specific to the Quadplane

For a quadplane operating at the autopilot-held cruise airspeed V_c , four balance equations connect the empirical measurements of Sec. VI to the airframe's design quantities. In level steady cruise, the lift balances the weight,

$$\frac{1}{2}\rho V_c^2 S C_L^{\text{cruise}} = W \implies C_L^{\text{cruise}} = \frac{2W}{\rho V_c^2 S}, \quad (18)$$

and the pusher thrust balances the drag,

$$T_{\text{pusher}} = D = \frac{1}{2}\rho V_c^2 S C_D^{\text{cruise}} \quad (19)$$

The corresponding electrical power required at the pusher motor, accounting for ESC and propeller efficiency, is

$$P_{\text{cruise,req}} = \frac{T_{\text{pusher}} V_c}{\eta_{\text{prop}} \eta_{\text{ESC}}} = \frac{\rho V_c^3 S C_D^{\text{cruise}}}{2 \eta_{\text{prop}} \eta_{\text{ESC}}} \quad (20)$$

Hence the measured cruise power $\bar{P}_{\text{cruise}}$ from Eq. (6) fixes the product $S C_D^{\text{cruise}}/(\eta_{\text{prop}}\eta_{\text{ESC}})$ and, given a nominal S , allows the aerodynamic and electromechanical loss contributions to be separated.

In hover, the four lift rotors collectively support the weight. The per-rotor ideal hover thrust, the disc loading, and the ideal induced power per rotor are

$$T_{\text{per}} = \frac{W}{N_r}, \quad \text{DL} = \frac{T_{\text{per}}}{A_d}, \quad P_{\text{ideal}}^{(1)} = T_{\text{per}} \sqrt{\frac{T_{\text{per}}}{2\rho A_d}}, \quad (21)$$

and the figure-of-merit relating ideal induced power to measured electrical power per rotor is

$$\text{FoM} = \frac{P_{\text{ideal}}^{(1)}}{P_{\text{measured}}^{(1)}} = \frac{T_{\text{per}} \sqrt{T_{\text{per}}/(2\rho A_d)}}{\bar{P}_{\text{hover}}/N_r} \quad (22)$$

A measured $\bar{P}_{\text{hover}}$ together with a known A_d therefore yields a configuration-level efficiency metric directly comparable across airframes.

In a coordinated banked turn at bank angle ϕ , the wing must support the load factor $n = 1/\cos \phi$ from Eq. (8), requiring an increased lift coefficient

$$C_L^{\text{turn}}(\phi) = \frac{2nW}{\rho V^2 S} = \frac{C_L^{\text{cruise}}}{\cos \phi} \quad (23)$$

The corresponding stall speed in the turn rises as

$$V_{\text{stall}}^{\text{turn}}(\phi) = V_{\text{stall,1g}} \sqrt{n} = V_{\text{stall,1g}} / \sqrt{\cos \phi}, \quad (24)$$

which bounds the maximum bank that can be sustained at the cruise airspeed.

For a head-on gust of magnitude Δv that reduces the freestream from V to $V - \Delta v$ at fixed angle of attack, the lift drops as the square of velocity,

$$\frac{\Delta L}{L} = \left(1 - \frac{\Delta v}{V}\right)^2 - 1 \approx -\frac{2\Delta v}{V} \quad \text{for } \Delta v \ll V \quad (25)$$

The corresponding airspeed margin for stall avoidance at the transition airspeed V_T is

$$\Delta v_{\max} = V_c - V_T, \quad (26)$$

so the operational head-wind ceiling satisfies $V_{\text{wind,max}} \lesssim \Delta v_{\max}$. The CFD case of Sec. VII.D provides the high- α correction to Eq. (25), where boundary-layer evolution amplifies the linear V^2 scaling.

D. Computational Aerodynamic Setup

A complementary CFD study is performed in Ansys Fluent (2022 R2 / 2023 R1) under an academic licence. The computational geometry is an untwisted, extruded S1223 wing panel of 0.292 m (11.5 in) chord and 1.016 m (40 in) span, matching the as-built airfoil templates released with the dataset; the panel planform area is 0.297 m² and its aspect ratio is 3.48, so the computed coefficients are finite-wing values rather than two-dimensional section values. The simulation is steady and incompressible, with a pressure-based solver and the SST $k-\omega$ turbulence model of Menter [35], which is appropriate for attached and mildly separated airfoil flow at moderate Reynolds number. The freestream condition is $V_\infty = 15$ m/s (the autopilot-held cruise airspeed; see Sec. VI), $\rho = 1.225$ kg/m³, and $\mu = 1.789 \times 10^{-5}$ Pa s, with turbulence intensity 1% at the inlet. Eq. (12) gives a chord Reynolds number of $\sim 3.0 \times 10^5$, in the low-Reynolds-number range typical of small UAV applications [32, 33]. In the chordwise plane, a symmetric C-grid mesh extends approximately 10 chord lengths upstream and 20 chord lengths downstream of the airfoil, with $y^+ \lesssim 1$ on the airfoil surface for boundary-layer resolution. A grid-refinement study comparing coarse, medium, and fine meshes at the cruise condition confirmed that the integrated lift and drag coefficients change by less than 1% between the medium and fine meshes; the medium mesh is used for the production angle-of-attack sweep. An angle-of-attack sweep is performed from $\alpha = -5^\circ$ to $+20^\circ$ in 5° increments, and lift and drag are extracted from CFD-Post via the global-direction force calculator on the wing surface; integrated forces are tabulated for $\alpha = -5^\circ$ to $+15^\circ$, while the $+20^\circ$ case exhibits the near-stall pressure topology and is used qualitatively. A separate airspeed-deficit case reduces the freestream from 15 to 13 m/s at $\alpha = +15^\circ$ as a quasi-steady estimate of gust-induced lift loss per Eq. (25); unsteady gust aerodynamics are not modelled, so the case is a conservative screening estimate. Convergence is monitored on the scaled residuals (each $< 10^{-5}$) and on the lift and drag force time histories.

IV. Hardware and Equipment

The test vehicle is a small quadplane VTOL UAV with a high-mounted, high-aspect-ratio main wing and an H-tail empennage on twin booms (Fig. 3). Four electrically driven lift rotors are mounted on the boom centrelines, one rotor at the front and one at the rear of each boom, giving a conventional “+”/“X” quadcopter arrangement when viewed from above. A separate brushless pusher motor with a folding propeller is mounted at the rear of the central fuselage and provides all thrust during forward flight. Control surfaces include differential ailerons on the main wing, an elevator on the horizontal stabilizer, and rudders on the twin vertical fins. The airframe is hand-built in fibreglass/composite skin over a foam core and is powered by a 6-cell lithium-polymer battery pack. The wing planform and airfoil cross-section are shown in Fig. 4; the airfoil is a thin, moderately cambered low-Reynolds-number section typical of small UAV applications [32, 33].

The principal system specifications are summarized in Table 3 and the equipment inventory in Table 4.

V. Experimental Setup

The flight analyzed in this paper was a ~24 min session conducted on 4 September 2022 in an open agricultural field near Dhaka, Bangladesh (23.85° N, 90.51° E, near sea level), of which ~6.5 min was armed (Fig. 5). Ambient conditions were tropical with hazy visibility and a moderate steady wind subsequently estimated via Eq. (5). The mission consisted of: 1) arming and pre-flight hover; 2) vertical climb to ~45 m AGL; 3) one complete MC→FW front transition; 4) ~4 min of fixed-wing cruise in a circular loiter pattern at ~45 m AGL and ~15.2 m/s airspeed; 5) one complete FW→MC back transition; and 6) vertical descent and recovery. A complete time-tagged log (log_5_2022-9-4-18-02-56.px4log) was recovered from the autopilot SD card; the per-topic CSV bundle is released alongside this paper (see Data Availability). An overview of the entire armed flight is given in Fig. 6, and the reconstructed three-dimensional flight path is shown in Fig. 7.

VI. Flight-Test Results

This section reports the results of the pipeline of Sec. III applied to the recorded flight log: hover (Sec. VI.A), transition (Sec. VI.B), power and energy (Sec. VI.C), lateral-directional cruise (Sec. VI.D), and longitudinal trim (Sec. VI.E). The aerodynamic interpretation appears in its own section (Sec. VII).

A. Hover-Phase Stability

Figure 8 shows p, q, r over a window spanning the last ~11 s of hover, the 4.5 s MC→FW transition, and the first ~9 s of cruise. The hover rates are tightly bounded:

$$\bar{p}_{\text{hover}} = +0.51 \pm 6.29 \text{ deg/s}, \quad \bar{q}_{\text{hover}} = -0.18 \pm 3.65 \text{ deg/s}, \quad \bar{r}_{\text{hover}} = -5.91 \pm 8.34 \text{ deg/s}.$$



Fig. 3 Test vehicle in operation: a)–f) ground and in-flight views; g) underside planform in steady cruise; h) side three-quarter view in late transition; i) low-altitude pass over the recovery area.

Mean roll and pitch rates are within sensor noise of zero; the small yaw rate is consistent with deliberate pre-transition heading alignment. The horizontal ground-speed magnitude V_h is essentially zero (0.16 ± 0.07 m/s), while the pitot reads $V_{\text{air}} = 2.93 \pm 0.34$ m/s (Fig. 9); Eq. (5) gives a wind component along body- x of ~ 2.77 m/s.

B. Hover-to-Forward-Flight Transition

The MC $\rightarrow$ FW transition begins when the autopilot toggles TransMode from 0 to 1 and ends when the lift rotors are commanded off. The pitot airspeed at each toggle defines the transition airspeed thresholds (Table 5, Fig. 10).

The front transition starts well below stall ($V \approx 3.83$ m/s), with the pusher motor spooling up while the lift rotors maintain support, and completes at $V_T = 11.92$ m/s ($\sim 28\%$ margin to cruise airspeed). The back-transition trigger

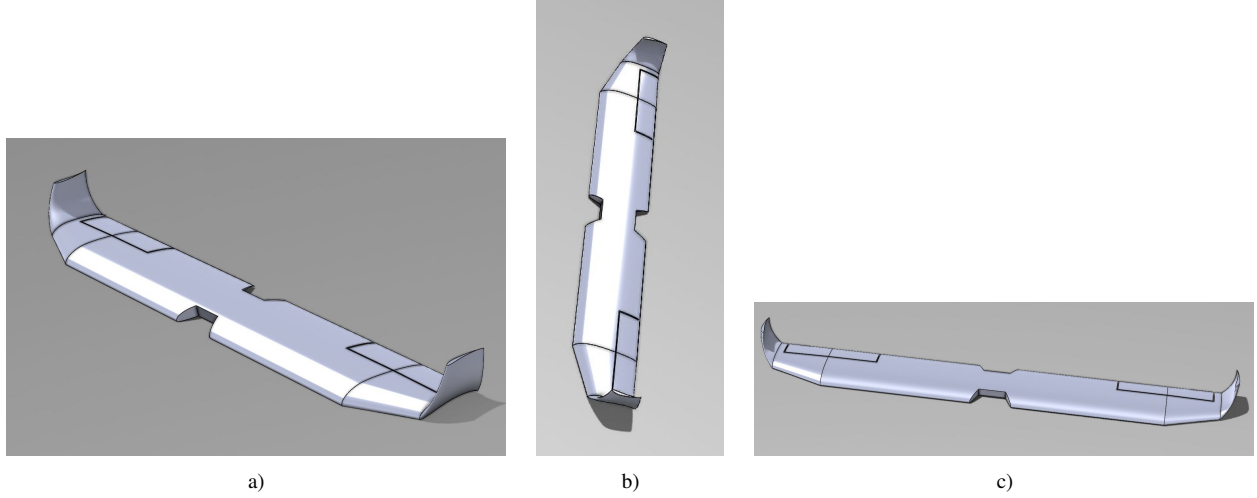


Fig. 4 Wing geometry: a) isometric CAD view; b) near end-on view showing thickness and tip-cant angle; c) lower perspective showing swept-up tip surfaces and control-surface outlines.



Fig. 5 Test environment: open field site, operator with ground-control station, and aircraft in low hover during recovery.

($V_{T,back} = 15.62 \text{ m/s}$) is slightly above cruise, so the lift rotors begin spooling up while the wing still has positive lift authority.

Pitch behaviour during transition (Fig. 11) is a primary indicator of how the airframe converts thrust into airspeed. From $t = 1156\text{--}1158 \text{ s}$, IAS climbs from 3.8 to $\sim 6 \text{ m/s}$ as the pusher spools up; pitch rises modestly to $+2^\circ$. Between 1158 and 1159 s the airspeed accelerates through the wing's low-speed regime and pitch peaks at $\theta_{max} = +5.40^\circ$. From 1159 to 1160.75 s the wing develops authority, the elevator actively trims, and pitch settles to the cruise mean of -0.57° . Transition body-rate peaks are $|p|_{max} \approx 55.5^\circ/\text{s}$, $|q|_{max} \approx 35.1^\circ/\text{s}$, $|r|_{max} \approx 35.0^\circ/\text{s}$ (Fig. 8, gold region), well below the PX4 default saturation limit ($\sim 220^\circ/\text{s}$). The back transition completes in 1.86 s with body rates $\leq 20^\circ/\text{s}$ throughout.

Table 3 System specifications of the test quadplane. Values marked [†] are estimated from imagery and component sizes; remaining values are derived from the flight log of Sec. V.

Category	Parameter	Value
Configuration	Aircraft class	Hybrid quadplane VTOL (lift+cruise)
	Lift propulsion	4× brushless DC, boom-mounted
	Cruise propulsion	1× brushless DC pusher (rear fuselage)
	Control surfaces	2 aileron, 1 elevator, 2 rudder
Geometry	Wing configuration	High-mounted, high aspect ratio
	Empennage	H-tail on twin booms
	Wingspan b	$\sim 2.0 \text{ m}^\dagger$
	Wing chord c	$\sim 0.25 \text{ m}^\dagger$
	Reference area S	$\sim 0.5 \text{ m}^2^\dagger$ (estimated from photogrammetry)
	Fuselage length	$\sim 1.7 \text{ m}^\dagger$
Mass / Loading	Gross mass m	$\sim 4 \text{ kg}$ to $6 \text{ kg}^\dagger$
	Wing loading W/S	$\sim 80 \text{ N/m}^2$ to $120 \text{ N/m}^2^\dagger$
	Hover disc loading $W/(N_r A_d)$	$\sim 30 \text{ N/m}^2$ to $50 \text{ N/m}^2^\dagger$
Performance (measured)	Cruise airspeed V_c	15.19(59) m/s
	Front-transition airspeed V_T	11.92 m/s
	Back-transition airspeed $V_{T,\text{back}}$	15.62 m/s
	Front-transition duration	4.53 s
	Back-transition duration	1.86 s
	Typical bank in turn	$ \phi \approx 25^\circ$
Electrical	Battery	6S LiPo, $\sim 10 \text{ A h}$, $\sim 222 \text{ W h}$
	Mean hover power	1269(24) W
	Mean cruise power	378(131) W
	Peak transition power	1597 W ($\sim 4 \text{ s}$ duration)
Endurance / Range	Pure-hover endurance (theoretical)	$\sim 10.5 \text{ min}$
	Pure-cruise endurance (theoretical)	$\sim 35 \text{ min}$
	Practical cruise range	$\sim 20 \text{ km}$
Autopilot	Flight stack	PX4 v1.7.0 (open-source)
	Log format	PX4 legacy sdlog2 (binary)

C. Power and Energy Budget

Phase-resolved electrical power is summarized in Table 6 and Fig. 12.

The hover-to-cruise power ratio is $\bar{P}_{\text{hover}}/\bar{P}_{\text{cruise}} = 1269/378 \approx 3.36$, the front-transition peak is 1597 W sustained for $\sim 4 \text{ s}$ (sets ESC/battery peak), and the avionics baseline is 97 W ($\sim 25\%$ of cruise). Applying Eq. (6) with $E_{\text{pack}} \approx 223 \text{ W h}$ gives a pure-hover endurance $\sim 10.5 \text{ min}$ and a pure-cruise endurance $\sim 35 \text{ min}$ (theoretical range $\sim 31 \text{ km}$). A practical mission with 30% pack reserve and $\sim 8 \text{ W h}$ hover overhead yields $\sim 20 \text{ km}$ of usable cruise range. Hover power is remarkably steady ($\sigma_P/\bar{P} \approx 1.9\%$). Applying Eq. (20) with the measured $\bar{P}_{\text{cruise}} \approx 378 \text{ W}$ and an assumed

Table 4 Equipment inventory of the test airframe. Items marked $\ddagger$ are typical of PX4-compatible builds of this class and reflect the as-built configuration; the published telemetry is independent of the specific manufacturer where not stated otherwise.

Subsystem	Component	Specification / Description
Avionics	Flight controller	Pixhawk-class, PX4-compatible flight controller
	IMU	MEMS 3-axis accelerometer + gyro $\ddagger$
	Magnetometer	3-axis digital magnetometer $\ddagger$
	Barometer	Digital pressure altimeter $\ddagger$
	GPS receiver	u-blox-class L1 with integrated compass $\ddagger$
	Airspeed sensor	Pitot tube + differential pressure transducer
Communications	Telemetry	SiK-compatible 433 MHz radio modem $\ddagger$
	RC receiver	SBUS / PPM $\ddagger$
	Ground station	QGroundControl (PX4 ecosystem)
Lift propulsion ($\times 4$)	Motor	Brushless DC outrunner, ~ 600 kV to 900 kV $\ddagger$
	Speed controller	BLHeli ESC, ≥ 25 A continuous $\ddagger$
	Propeller	Fixed-pitch, ~ 10 – 12 in diameter $\ddagger$
Cruise propulsion ($\times 1$)	Motor	Brushless DC, ~ 700 kV to 900 kV $\ddagger$
	Speed controller	BLHeli ESC, ≥ 30 A continuous $\ddagger$
	Propeller	Folding, ~ 10 – 13 in diameter $\ddagger$
Actuation	Aileron servos ($\times 2$)	Standard hobby-grade, ~ 0.10 s/ 60° $\ddagger$
	Elevator servo ($\times 1$)	Standard hobby-grade $\ddagger$
	Rudder servos ($\times 2$)	Standard hobby-grade $\ddagger$
Power	Battery pack	6S LiPo, nominal 22.2 V, ~ 10 A h
	Power module	PX4-compatible $V+I$ sensor, in-line
Airframe	Wing	Fibreglass/composite skin over foam core
	Fuselage	Fibreglass/composite shell
	Booms	Twin carbon-fibre booms with $4\times$ lift rotors

Table 5 Transition airspeed thresholds and durations from the TransMode flag toggles.

Event	Time, s	IAS, m/s
Front-transition commit	1156.22	3.83
Front-transition complete	1160.75	11.92 ($= V_T$)
Back-transition commit	1407.32	15.62 ($= V_{T,back}$)
Back-transition complete	1409.18	12.88
<i>Front-transition duration</i>	—	4.53 s
<i>Back-transition duration</i>	—	1.86 s

$\eta_{\text{prop}}\eta_{\text{ESC}} \approx 0.45$ gives an aerodynamic $T_{\text{pusher}} \approx 11.3$ N and a lift-to-drag ratio at cruise $\approx W/T_{\text{pusher}}$ in the range 3.5–5.0 for the 4–6 kg mass band—roughly half the clean-panel maximum of ~ 10 from the Fluent polar (Sec. VII.B),

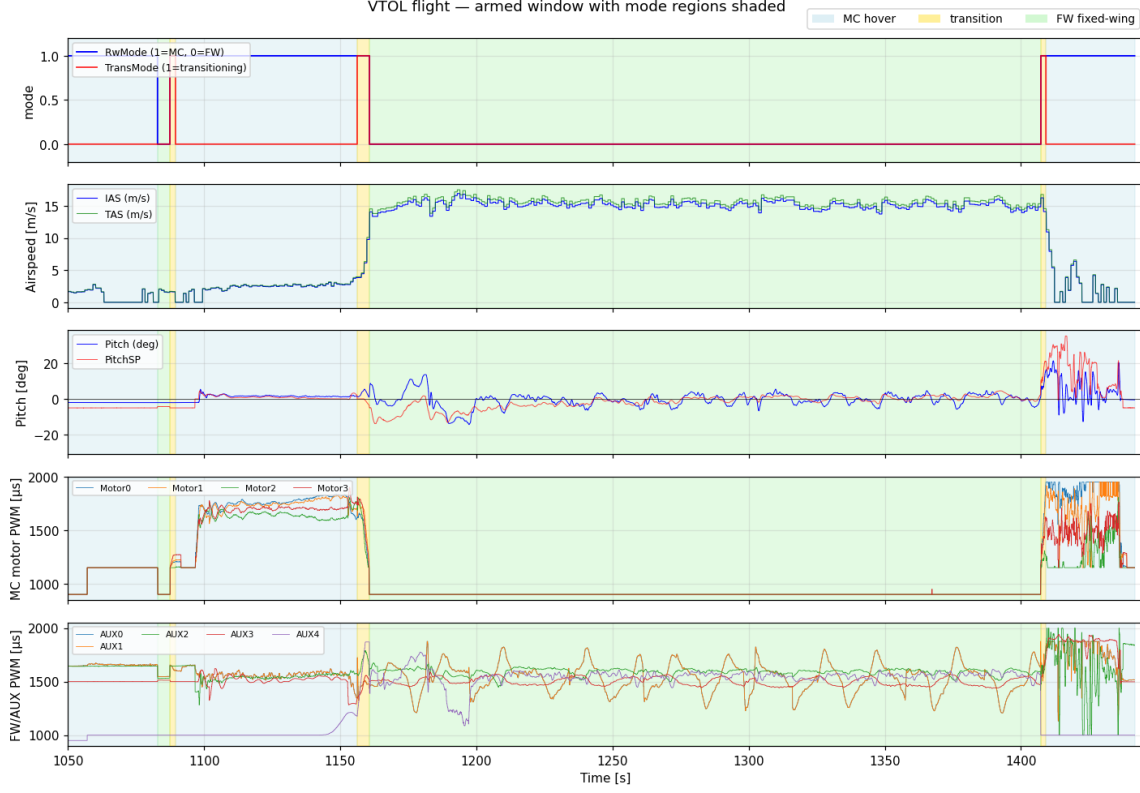


Fig. 6 Armed-flight overview: mode flags, airspeed, pitch and setpoint, and lift-motor and pusher PWM commands, with flight phases shaded.

Table 6 Phase-resolved electrical power and energy.

Phase	Duration, s	Mean P , W	Peak P , W	Energy, W·h
Pre-takeoff (avionics)	27	97 ± 12	130	0.73
Climb-out (MC)	50	1010 ± 407	1244	14.01
Stable hover (MC)	11	1269 ± 24	1333	3.92
Front transition (MC→FW)	4.5	1368 ± 146	1597	1.73
Cruise (FW)	235	378 ± 131	1010	24.81
Back transition (FW→MC)	1.9	380 ± 13	393	0.20
Landing (MC)	31	890 ± 253	1245	7.71

the deficit reflecting fuselage, boom, stopped-rotor, and trim drag absent from the panel model. All weight-derived quantities inherit the 4–6 kg mass-estimate uncertainty and are therefore quoted as ranges. The measured pack-level power profile sets the requirements on the on-board energy-management layer, for which data-driven state-of-charge estimation and balancing schemes [36, 37] are an active research direction.

D. Lateral-Directional Cruise Behaviour

The 235 s cruise window is a continuous circular loiter at ~ 45 m AGL. The autopilot holds IAS to 15.19 ± 0.59 m/s (4% of setpoint) and altitude to ± 2 m. Figure 13 shows the bank-angle trace and distribution: the aircraft spends 83.8% of

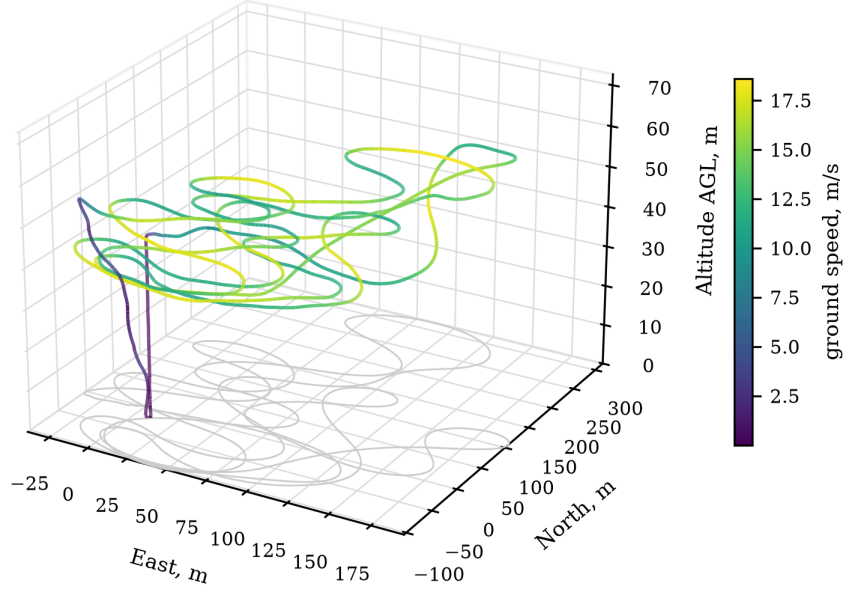


Fig. 7 Three-dimensional flight path reconstructed from the local-position estimator, coloured by ground speed; the gray curve is the ground-track projection.

cruise turning ($|\phi| > 10^\circ$), with mean right-turn bank $+27.1^\circ$, mean left-turn bank -23.1° , and peaks of $+53.8^\circ/-45.6^\circ$. The mean signed bank $+1.03^\circ$ confirms left/right cancellation over the loiter. The turn radius is computed two ways via Eq. (8): the aerodynamic formula and the kinematic $R = V_h/|\dot{\psi}|$. Median radii of $R_{\text{formula}} = 54$ m and $R_{\text{kinematic}} = 48$ m agree to $\sim 10\%$, confirming coordinated turning. Typical operating radius is ~ 50 m, with tight ~ 20 m turns at maximum bank. Eq. (23) requires $C_L^{\text{turn}}/C_L^{\text{cruise}} = 1/\cos 25^\circ \approx 1.10$ at the typical bank; from Fig. 16, this remains well within the linear regime.

E. Longitudinal Trim During Cruise

Figure 14 shows the elevator PWM command during cruise, with red dots marking samples satisfying the level-flight criterion ($|\phi| < 5^\circ$, $|\dot{h}| < 0.5$ m/s; $n = 65$ samples). The level-cruise mean is $1607 \mu\text{s}$ with $\sigma = 11.9 \mu\text{s}$, an offset of $+107 \mu\text{s}$ from neutral. Applying Eq. (9) with $\delta_{\text{max}} = 30^\circ$ yields $|\delta_e^{\text{trim}}| \approx 6.4^\circ$ with sub-degree jitter, consistent with a well-tuned pitch-attitude loop.

VII. Computational Aerodynamic Analysis

To interpret the empirical V_T threshold and quantify the wing's sensitivity to airspeed disturbance, an angle-of-attack sweep on the wing airfoil was performed in Ansys Fluent following the setup of Sec. III.D. This section reports the multi-AoA pressure field (Sec. VII.A), the lift and drag polars (Sec. VII.B), qualitative 3D flow visualization (Sec. VII.C),

Body angular rates p (roll), q (pitch), r (yaw): ≈ 0 in stable hover, large excursions during transition



Fig. 8 Body-axis angular rates across hover, transition, and fixed-wing flight, comparing gyro rates with numerical Euler-angle derivatives.

and the airspeed-deficit case (Sec. VII.D). Implications for V_T and the wind ceiling are summarized in Sec. VII.E.

A. Multi-Angle Pressure Field

Figure 15 shows the mid-span static pressure contour around the wing section at $\alpha = -5^\circ, +5^\circ, +15^\circ, +20^\circ$, $V_\infty = 15$ m/s. At $\alpha = -5^\circ$ the pressure field is nearly fore-aft symmetric with weak upper-surface low pressure, an essentially zero-lift condition for the cambered section. At $\alpha = +5^\circ$ a clear low-pressure suction peak develops on the upper surface near the leading edge and a higher-pressure plateau forms on the lower surface, the canonical attached-flow configuration. At $\alpha = +15^\circ$ the suction peak intensifies and migrates toward the leading edge. At $\alpha = +20^\circ$ the suction peak concentrates within a small leading-edge region while the lower-surface pressure plateau is at its strongest; the topology signals that the upper-surface boundary layer is on the edge of separation.

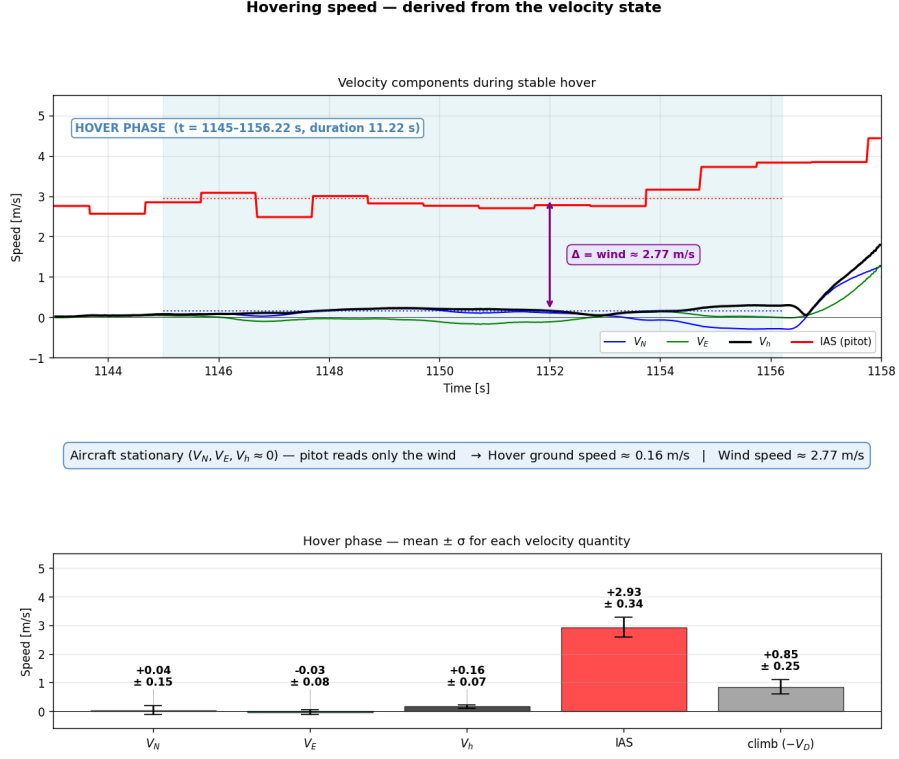


Fig. 9 Velocity components during stable hover; the pitot–ground-speed difference is the steady wind along body- x .

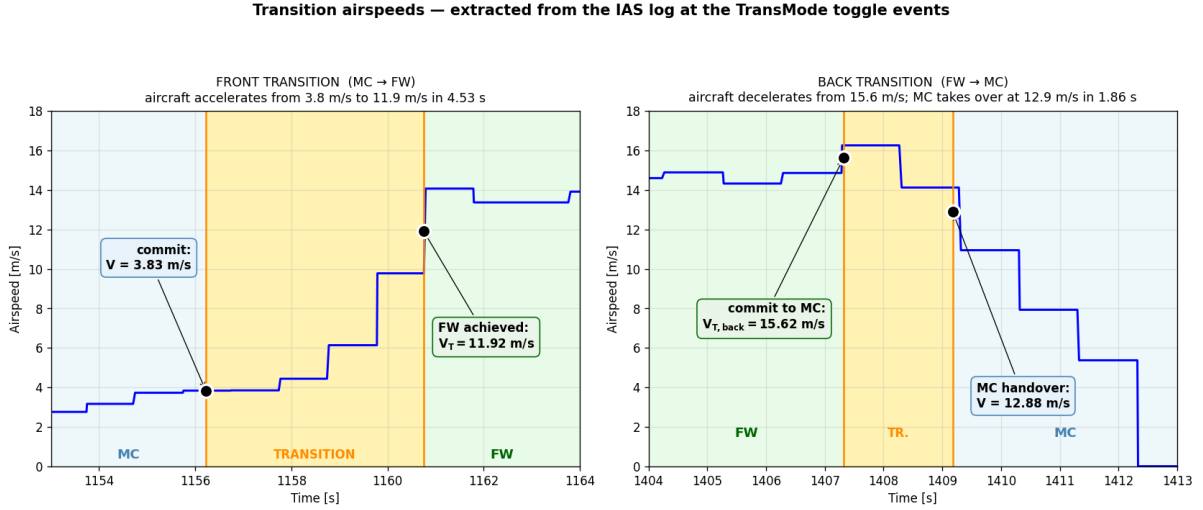


Fig. 10 Pitot IAS through the front and back transitions, with TransMode toggles marked.

B. Lift and Drag Polars

Figure 16 reports the wing-panel $C_L(\alpha)$ and $C_D(\alpha)$, obtained by normalizing the integrated Fluent forces by the panel planform area ($S_{\text{panel}} = 0.297 \text{ m}^2$) and freestream dynamic pressure. The lift coefficient rises essentially linearly from 0.31 at $\alpha = -5^\circ$ to 1.23 at $+15^\circ$, with fitted lift-curve slope $C_{L_\alpha} \approx 2.64/\text{rad}$ (0.046/deg) and zero-lift angle

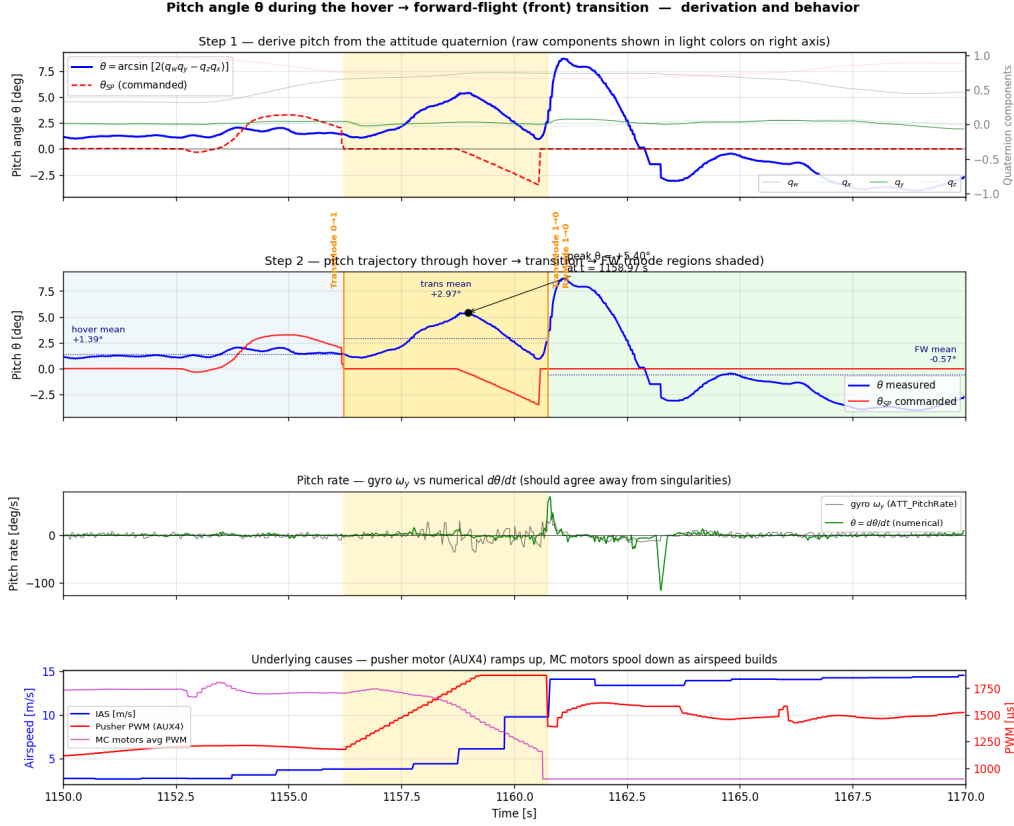


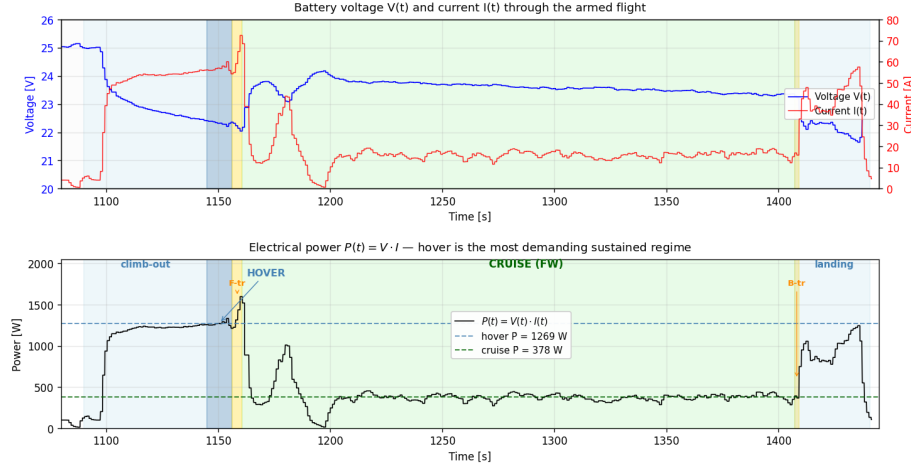
Fig. 11 Pitch behaviour during front transition: quaternion derivation, pitch trajectory with phase means, pitch rate, and pusher spool-up.

$\alpha_{0L} \approx -12^\circ$; the strongly negative zero-lift angle reflects the high camber of the S1223 section. The finite aspect ratio of the panel accounts for most of the reduction from the two-dimensional section slope: lifting-line theory with a section slope of $\sim 5.7/\text{rad}$ predicts a wing slope of $3.4\text{--}3.7/\text{rad}$ at aspect ratio 3.48, with the residual gap attributable to low-Reynolds-number viscous decambering [32–34]. The incremental lift per 5° step decreases monotonically toward $+15^\circ$, and the $+20^\circ$ pressure topology of Fig. 15 indicates incipient separation, placing stall onset in the $+15^\circ\text{--}+20^\circ$ range. The drag data provide a quantitative validation of the parabolic polar of Eq. (11): a linear fit of C_D against C_L^2 [Fig. 16c] yields $C_{D0} = 0.018$ and $k = 0.120$, corresponding to an Oswald efficiency $e = 1/(\pi \text{AR} k) = 0.77$ —squarely in the expected range for an untwisted rectangular panel. The panel’s maximum lift-to-drag ratio is ~ 10.4 , attained near $\alpha = 0^\circ$.

C. Three-Dimensional Flow Visualization

Qualitative 3D flow visualization (Ansys Discovery, lower-fidelity preview solver) is shown in Fig. 17. Streamlines and velocity vectors confirm largely attached flow on both surfaces, with mild streamline curvature in the immediate vicinity of the wing. The 3D views also indicate that spanwise flow is small under symmetric inflow, consistent with the

Hover power consumption — $P = V \cdot I$ from battery telemetry



Hover power derivation: $P_{\text{hover}} = V \cdot I = 22.33 \text{ V} \times 56.84 \text{ A} = 1269 \text{ W}$ | Energy over 11.2 s: $E = \int P dt = 14126 \text{ J} = 3.92 \text{ Wh}$ | Endurance on ~223 Wh pack: **10.5 min**

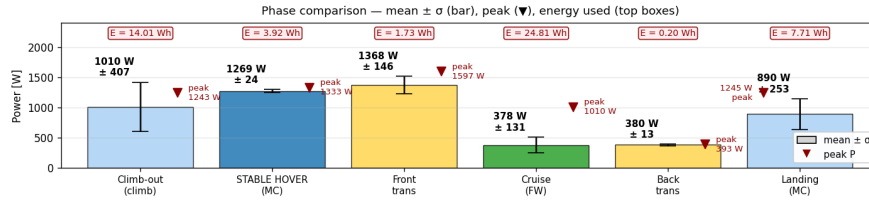


Fig. 12 Electrical power budget: pack voltage and current, instantaneous power with phase means, and per-phase energy.

near-uniform spanwise loading assumed in interpreting the panel polars. Higher-fidelity 3D simulations are reserved for future work.

D. Airspeed-Deficit (Quasi-Steady Gust) Case

A separate Fluent case reduces the freestream from 15 to 13 m/s at $\alpha = +15^\circ$, a quasi-steady representation of a head-on gust encountered while the wing operates at high-angle-of-attack transition margin; unsteady gust dynamics are not modelled, so the result is a conservative screening estimate. The integrated lift drops from 50.4 N to 33.8 N, a ~33% reduction (Fig. 18). Quadratic dynamic-pressure scaling [Eq. (25)] accounts for $1 - (13/15)^2 \approx 25\%$; the remaining ~8 percentage points correspond to a 10.7% reduction in lift coefficient (from 1.23 to 1.10, the open marker in Fig. 16a), reflecting boundary-layer degradation on the suction side near the stall edge, consistent with the leading-edge-concentrated topology of Fig. 15. This case anchors the operational wind ceiling derived in Sec. VIII.E.

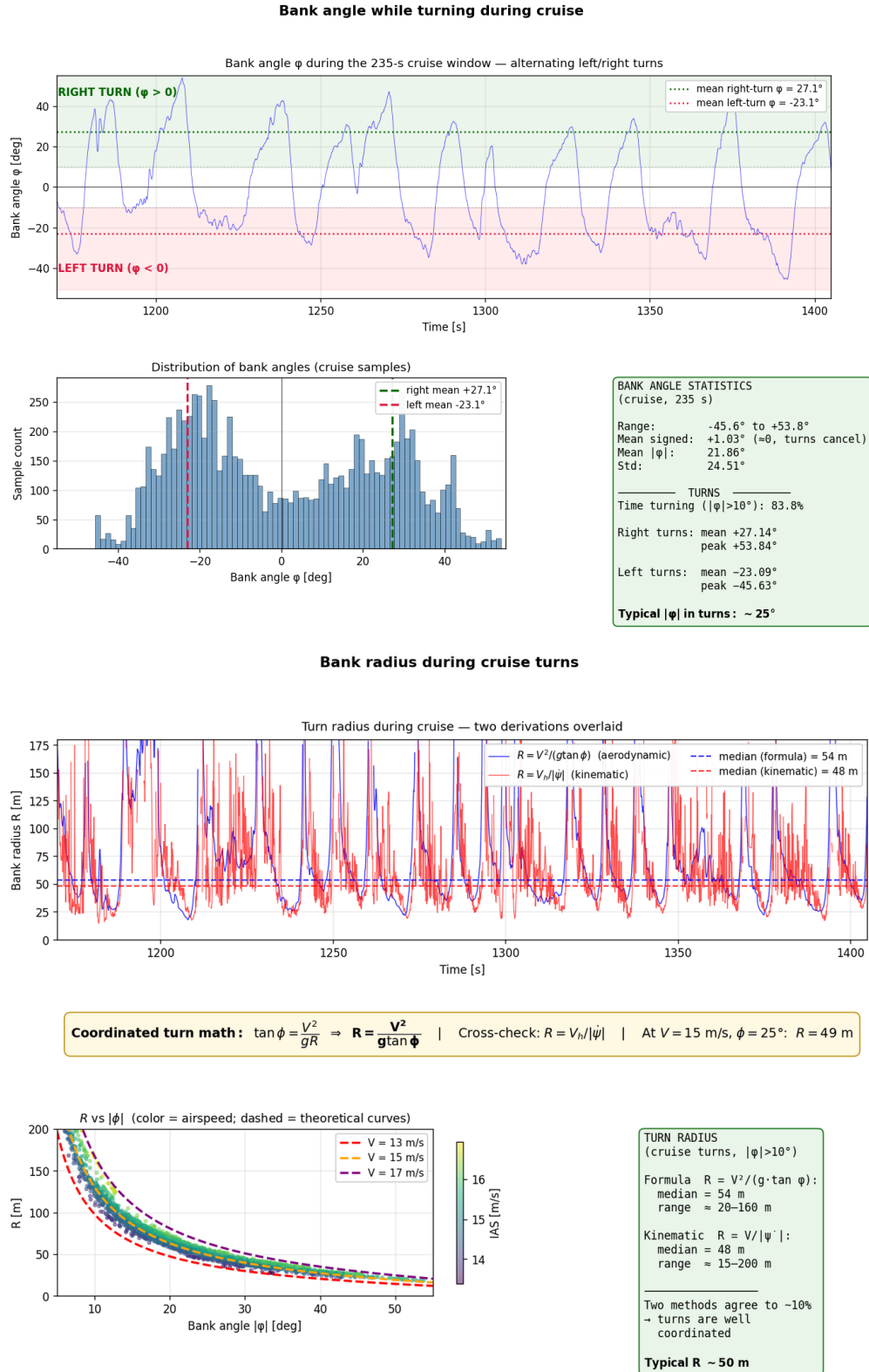


Fig. 13 Lateral-directional cruise: bank-angle trace and distribution, and turn radius by aerodynamic and kinematic estimates.

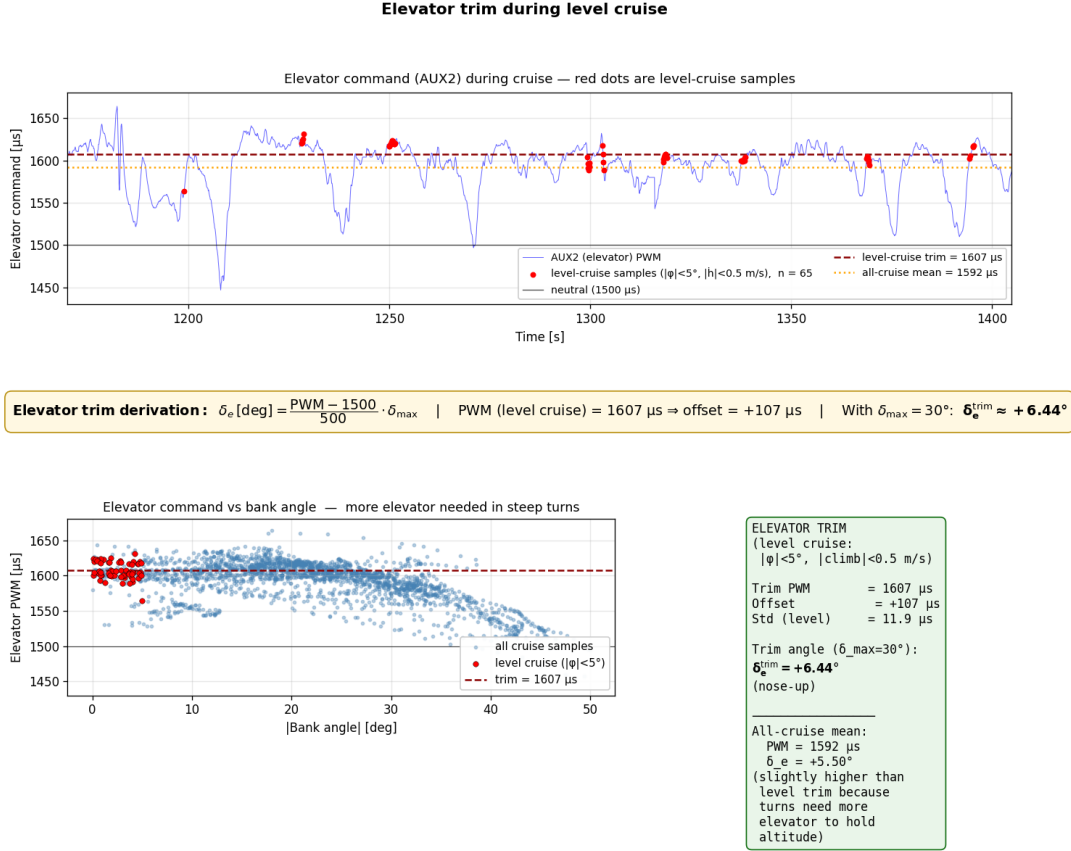


Fig. 14 Elevator command during cruise with level-cruise samples marked; trim deflection is approximately 6.4° .

E. Aerodynamic Interpretation of the Transition Airspeed

With the estimated aircraft wing area ($S \approx 0.5 \text{ m}^2$) and the 4–6 kg mass band, the lift balance of Eq. (18) requires $C_L = 0.56\text{--}0.83$ in cruise at 15.2 m/s, rising by the factor $(15.19/11.92)^2 \approx 1.62$ to $C_L = 0.90\text{--}1.35$ at $V_T = 11.92 \text{ m/s}$. The lower half of that requirement sits comfortably in the linear regime of the panel polar [Fig. 16a)]; toward the 6 kg end it approaches the highest lift coefficient demonstrated by the panel (1.23 at $+15^\circ$). The flight wing, at aspect ratio ~ 8 , has a steeper lift-curve slope and a higher maximum lift coefficient than the aspect-ratio-3.48 test panel, so the panel polar is a conservative surrogate; nevertheless, the thinning margin toward the heavy end of the mass estimate underlines why V_T cannot be reduced. An increase in V_T , conversely, would waste energy on prolonged transition. The airspeed-deficit case (Sec. VII.D) shows that a modest 2 m/s disturbance consumes much of this margin at high angle of attack, providing the quantitative basis for the wind-tolerance rule of Sec. VIII.E.

VIII. Design Implications

The measured flight characteristics and CFD interpretation of V_T motivate several design rules for quadplane-class VTOL UAVs, stated in inequality form for reusability, with the caveat that the present results derive from a single flight

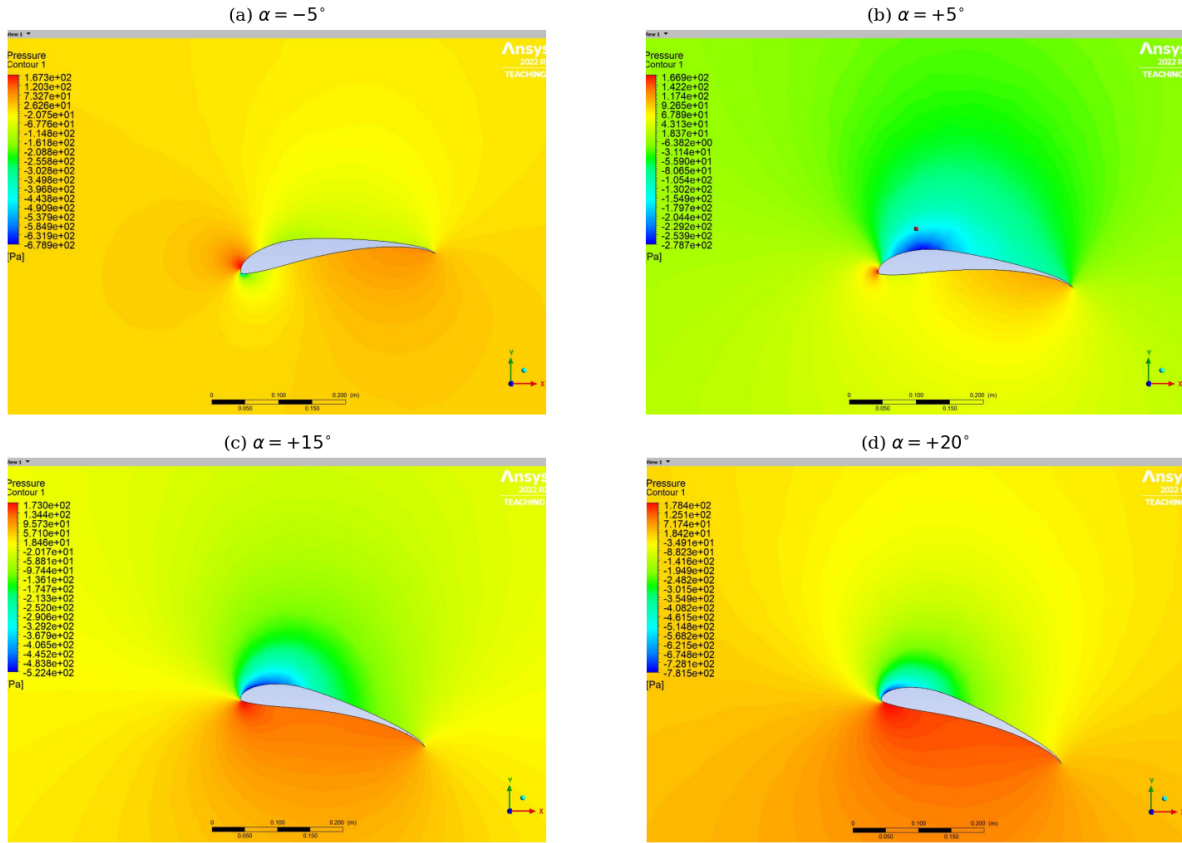


Fig. 15 Static pressure contours at four angles of attack, from near-symmetric flow at -5° to near-stall topology at $+20^\circ$.

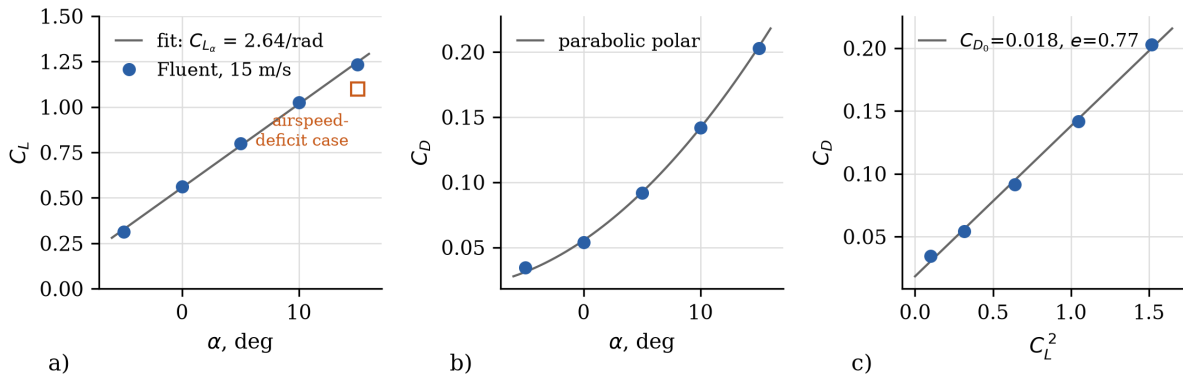


Fig. 16 Wing-panel polars: a) lift coefficient with linear fit and the airspeed-deficit case; b) drag coefficient with parabolic-polar fit; c) drag versus lift-squared, yielding $C_{D0} = 0.018$ and Oswald efficiency $e = 0.77$.

on one airframe; broader statistical validation is future work.

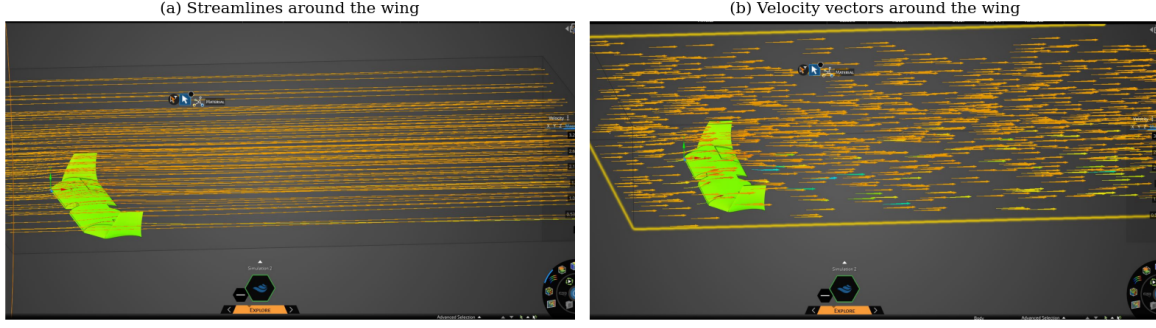


Fig. 17 Preliminary 3D flow visualization: a) streamlines showing largely attached flow; b) predominantly streamwise velocity vectors.

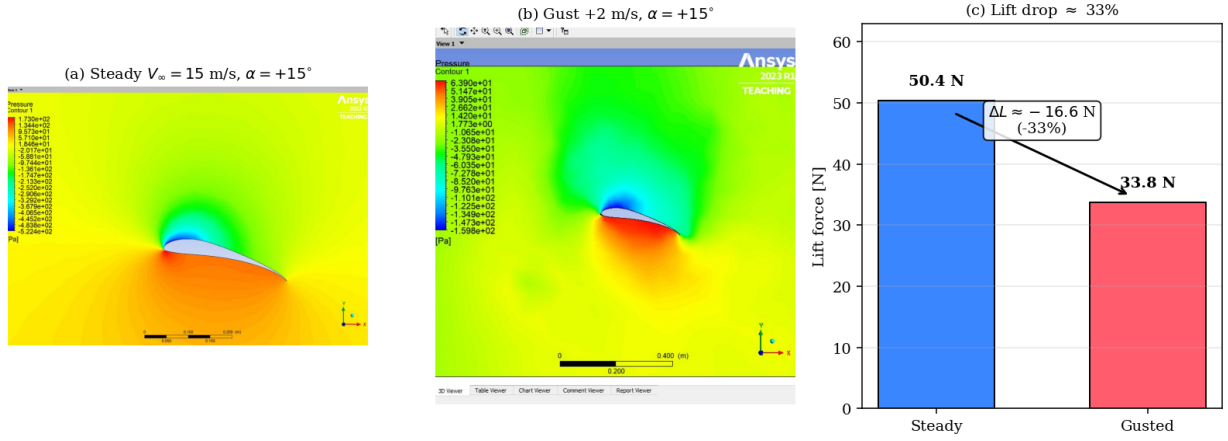


Fig. 18 Airspeed-deficit case at $\alpha = +15^\circ$: a) steady and b) reduced-freestream pressure contours; c) integrated lift drops by approximately 33%, of which 25% is dynamic-pressure scaling and the remainder boundary-layer degradation.

A. Transition Airspeed Margin

With $V_T = 11.92$ m/s sitting $\sim 28\%$ below cruise (15.2 m/s), the CFD lift curve places the wing well within the linear $C_L(\alpha)$ regime at V_T . The measurement and CFD together support

$$V_{\text{cruise}} \geq 1.25 V_T \quad \text{and} \quad V_{T,\text{back}} \geq V_{\text{cruise}} \quad (27)$$

B. Control Authority Budget

Peak transition body rates of $\sim 55^\circ/\text{s}$ (roll) and $\sim 35^\circ/\text{s}$ (pitch/yaw) sit well below the PX4 default $220^\circ/\text{s}$ limits, giving more than $4\times$ headroom:

$$\dot{\Phi}_{\text{controller,max}} \geq 4 \dot{\Phi}_{\text{trans,peak}} \quad (28)$$

C. Power-System Sizing

Hover (~ 1.3 kW sustained) and front transition (1.6 kW transient) define worst-case discharge. For a 223 Wh $6S$ pack (~ 10 Ah at 22 V), continuous 60 A implies $6C$ and peak 72 A implies $7.2C$ —within commodity $6S$ LiPo specifications. Per-rotor hover current of ~ 15 A indicates ≥ 25 A ESCs are adequate.

D. Energy Budget and Range

With Eq. (7), missions with $\eta_{\text{hover}} \leq 0.10$ are cruise-dominated and approach the ~ 35 min pure-cruise endurance. With 30% pack reserve and ~ 8 Wh hover overhead, practical cruise range is ~ 20 km. The avionics baseline of 97 W ($\sim 25\%$ of cruise) is itself a candidate for reduction.

E. Operational Wind Ceiling

The quasi-steady airspeed-deficit case (Sec. VII.D) shows a 2 m/s airspeed disturbance at $\alpha = +15^\circ$ removes $\sim 33\%$ of the lift. With Eq. (26), a conservative operational ceiling is

$$V_{\text{wind,max}} \lesssim V_{\text{cruise}} - V_T \approx 3.3 \text{ m/s}, \quad (29)$$

or larger only with confirmed gust-rejection authority. Adaptive transition strategies in gust-heavy environments are an active research direction [21].

F. Configuration Choice

The lift+cruise layout offers two advantages consistent with comparative analyses [1, 2, 8]: mode independence (lift rotors decoupled from pusher and wing, eliminating tilt-mechanism complexity [16, 18]) and clean stall margins (all four lift rotors idle in cruise). The trade-off is that the lift rotors and booms are parasitic in cruise, observable in the cruise/hover ratio.

IX. Conclusions

Flight-test telemetry from a small quadplane VTOL UAV, recorded by an open-source PX4 autopilot in the legacy sdlog2 format, has been combined with an Ansys Fluent angle-of-attack sweep of a finite-span S1223 wing panel to characterize the hover phase, the MC $\rightarrow$ FW transition, the FW cruise, and the FW $\rightarrow$ MC back transition. The empirical pipeline is reproducible from the binary log; the CFD provides the physical interpretation of the empirical transition airspeed. A comparison table (Table 2) confirms that no prior study covers the same combination of axes (configuration, methodology, data source, transition-airspeed reporting, transition-duration reporting, power-budget reporting, lateral-directional characterization, and CFD aerodynamic interpretation) on a quadplane platform.

In hover, body rates are within sensor noise ($|\bar{p}|, |\bar{q}| < 0.5^\circ/\text{s}$) and pitot recovers the steady wind (2.77 m/s). The

MC→FW transition completes in 4.53 s at $V_T = 11.92$ m/s with pitch bounded at $+5.4^\circ$ and body rates peaking at $\sim 55^\circ/\text{s}$; the back transition completes in 1.86 s. Cruise airspeed is held within 4% of setpoint with $\sim 25^\circ$ banks in coordinated turns of ~ 50 m radius, two derivations agreeing to 10%. Elevator trim is $\delta_e^{\text{trim}} \approx +6.4^\circ$. The hover-to-cruise power ratio is 3.36 $\times$; transition surges to 1.6 kW.

The computational sweep provides finite-panel lift and drag polars (lift-curve slope 2.64/rad, zero-lift angle -12° , zero-lift drag coefficient 0.018, Oswald efficiency 0.77), identifies stall onset above $+15^\circ$, and shows that a 2 m/s quasi-steady airspeed deficit removes $\sim 33\%$ of the lift—supplying the quantitative basis for the operational wind ceiling. Design implications follow directly from the combined evidence.

The results derive from a single flight on one airframe; broader statistical validation through additional flights and platforms is required and is the central theme of future work. Planned extensions include higher-fidelity 3D CFD with confirmed reference area and mass, IMU vibration spectra during cruise vs. hover, per-motor current isolation, and disc-loading/figure-of-merit estimation once rotor geometry is measured directly. Further methodological extensions include the integration of safe model-free reinforcement-learning controllers with state and control constraints via control barrier functions and discrete-event safety supervision over the autopilot’s mode logic, building on the methods reviewed in the introduction.

Data Availability

The analysis pipeline operates on per-topic CSV exports of the original PX4 sdlog2 binary log (log_5_2022-9-4-18-02-56.px4log, 4 September 2022). Thirteen CSV topic files (ATT, ATSP, ARSP, AIRS, BATT, GPOS, GPS, LPOS, LPSP, OUT0, PWR, RC, STAT), together with a column-level data dictionary, the Ansys CFD and structural-FEA project files, the S1223 airfoil geometry, the CAD models, and the Python analysis code used to generate flight figures, are released at the public GitHub repository <https://github.com/AdamDony/quadplane-px4-flight-data>. The VTOL mode-flag topic (VTOL) and auxiliary PWM bank (OUT1) did not survive log archival and are not part of the release; the transition events derived from them are reported in Table 5, and the transition window is independently visible in the released OUT0 lift-motor channels. An archival snapshot will be deposited on Zenodo, and the DOI will be inserted upon acceptance.

Funding Sources

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Acknowledgments

The authors gratefully acknowledge the PX4 open-source autopilot project and its developer community for the telemetry framework that made this analysis possible. An AI assistant (Claude, Anthropic) was used to assist with the organization of the public data repository and the development of the data-analysis and figure-preparation code; all analyses, results, and text were reviewed and validated by the authors, who take full responsibility for the content of this article.

References

- [1] Saeed, A. S., Younes, A. B., Cai, C., and Cai, G., “A Survey of Hybrid Unmanned Aerial Vehicles,” *Progress in Aerospace Sciences*, Vol. 98, 2018, pp. 91–105. <https://doi.org/10.1016/j.paerosci.2018.03.007>
- [2] Osman, M., Aboul-Magd, M. M., and Shaheen, A. A., “A Comprehensive Review of Hybrid VTOL UAV Technologies: Configurations, Propulsion, and Applications,” *Alexandria Engineering Journal*, 2025.
- [3] Rahman, T., and Islam, M. A., “A Complete Design and Implementation of a Low Cost Remote Controlled Flying Machine with FPGA-Based Control System,” *Proceedings of the 2nd International Conference on Computer and Automation Engineering (ICCAE)*, IEEE, Singapore, 2010, pp. 771–775. <https://doi.org/10.1109/ICCAE.2010.5451246>
- [4] Hossain, M. S., Kabir, A. M., Mazumder, P., Aziz, A., Hassan, M., and Islam, M. A., “Design and Development of a Y4 Copter Control System,” *Proceedings of the UKSim 14th International Conference on Computer Modelling and Simulation*, IEEE, 2012, pp. 12–17.
- [5] Islam, M. A., Rodriguez, J., Saha, A. K., Hwang, Y., and Jo, B. W., “Design and Implementation of a Variable Camber Mechanism for a Small Fixed-Wing UAV,” *Proceedings of the 2024 IEEE International Conference on Robotics and Biomimetics (ROBIO)*, IEEE, 2024, pp. 1676–1681.
- [6] Islam, M. A., Rahman, A., Saha, A., and Jo, B. W., “Design, Implementation, and Test Flight of a Lightweight Camber Morphing Wing Aircraft,” *AIAA Aviation Forum and Ascend*, AIAA Paper 2025-3524, 2025.
- [7] Islam, M. A., Rahman, A., Saha, A., and Jo, B. W., “Design of a 3D-Printed and Modular Camber Morphing Wing Mechanism and Its Performance Analysis in a Wind Tunnel,” *Proceedings of the ASME 2025 International Mechanical Engineering Congress and Exposition (IMECE)*, Vol. 3, Memphis, TN, 2025, Art. V003T06A023. <https://doi.org/10.1115/IMECE2025-166937>
- [8] Bacchini, A., and Cestino, E., “Electric VTOL Configurations Comparison,” *Aerospace*, Vol. 6, No. 3, 2019, Paper 26. <https://doi.org/10.3390/aerospace6030026>
- [9] Maqsood, A., and Go, T. H., “Optimization of Transition Maneuvers Through Aerodynamic Vectoring,” *Aerospace Science and Technology*, Vol. 23, No. 1, 2012, pp. 363–371. <https://doi.org/10.1016/j.ast.2011.09.004>

- [10] Liu, Z., He, Y., Yang, L., and Han, J., “Control Techniques of Tilt Rotor Unmanned Aerial Vehicle Systems: A Review,” *International Journal of Aerospace Engineering*, Vol. 2018, 2018, Paper 6315856. <https://doi.org/10.1155/2018/6315856>
- [11] Beach, J. M., Argyle, M. E., McLain, T. W., Beard, R. W., and Morris, S., “Tailsitter Attitude Control Using Resolved Tilt-Twist,” *Proceedings of the 2014 International Conference on Unmanned Aircraft Systems (ICUAS)*, IEEE, Orlando, FL, 2014, pp. 768–779. <https://doi.org/10.1109/ICUAS.2014.6842322>
- [12] Lyu, X., Gu, H., Zhou, J., Li, Z., Shen, S., and Zhang, F., “A Hierarchical Control Approach for a Quadrotor Tail-Sitter VTOL UAV and Experimental Verification,” *Proceedings of the 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, Vancouver, Canada, 2017, pp. 5135–5141. <https://doi.org/10.1109/IROS.2017.8206400>
- [13] Verling, S., Weibel, B., Boosfeld, M., Alexis, K., Burri, M., and Siegwart, R., “Full Attitude Control of a VTOL Tailsitter UAV,” *Proceedings of the 2016 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, Stockholm, Sweden, 2016, pp. 3006–3012. <https://doi.org/10.1109/ICRA.2016.7487466>
- [14] Kubo, D., and Suzuki, S., “Tail-Sitter Vertical Takeoff and Landing Unmanned Aerial Vehicle: Transitional Flight Analysis,” *Journal of Aircraft*, Vol. 45, No. 1, 2008, pp. 292–297. <https://doi.org/10.2514/1.30122>
- [15] Ritz, R., and D’Andrea, R., “A Global Controller for Flying Wing Tailsitter Vehicles,” *Proceedings of the 2017 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, Singapore, 2017, pp. 2731–2738. <https://doi.org/10.1109/ICRA.2017.7989318>
- [16] Geuther, S. C., North, D. D., and Busan, R. C., “Investigation of a Tandem Tilt-Wing VTOL Aircraft in a Sub-Scale Free-Flight Test,” NASA TM-2020-5003178, NASA Langley Research Center, Hampton, VA, 2020.
- [17] Yeh, Y. L., Yan, X., and Du, S., “Inverse Machine Learning Framework for Optimizing the Takeoff Performance of a Tilt-Wing eVTOL Vehicle,” *AIAA Aviation 2023 Forum*, AIAA Paper 2023-3593, San Diego, CA, 2023. <https://doi.org/10.2514/6.2023-3593>
- [18] May, M. S., Milz, D., and Looye, G., “Transition Strategies for Tilt-Wing Aircraft,” *AIAA SciTech 2024 Forum*, AIAA Paper 2024-1289, Orlando, FL, 2024. <https://doi.org/10.2514/6.2024-1289>
- [19] Mathur, A., and Atkins, E. M., “Experimental Aerodynamic Analysis of a QuadPlane Unmanned Aircraft System,” *Journal of Aircraft*, Vol. 60, No. 4, 2023, pp. 1323–1328. <https://doi.org/10.2514/1.C036797>
- [20] Lee, S., and Kim, S., “Model Predictive Control-Based Longitudinal Control Design for a Fixed-Wing VTOL UAV,” *AIAA SciTech 2024 Forum*, AIAA Paper 2024-0141, Orlando, FL, 2024. <https://doi.org/10.2514/6.2024-0141>
- [21] Chiwande, S., “INAV-Based Adaptive Transition Control for QuadPlane VTOL UAVs in Urban Wind Gust Environments,” *SSRG International Journal of Electrical and Electronics Engineering*, Vol. 12, No. 8, 2025, pp. 186–195.

- [22] Dong, W., Dony, M. N.-A., and Rafat, M., “Tracking Control of Vertical Take-Off and Landing Vehicles with Parametric Uncertainty,” *International Journal of Adaptive Control and Signal Processing*, Vol. 34, 2020. <https://doi.org/10.1002/acs.3143>
- [23] Dony, M. N.-A.-A., and Dong, W., “Distributed Robust Formation Flying and Attitude Synchronization of Spacecraft,” *Journal of Aerospace Engineering*, Vol. 34, No. 3, 2021. [https://doi.org/10.1061/\(ASCE\)AS.1943-5525.0001262](https://doi.org/10.1061/(ASCE)AS.1943-5525.0001262)
- [24] Dony, M. N.-A.-A., “Robust Distributed Formation Control of UAVs with Higher-Order Dynamics,” Ph.D. Dissertation, The University of Texas Rio Grande Valley, Edinburg, TX, 2021.
- [25] Dony, M. N.-A.-A., Rafat, M., and Dong, W., “Distributed Command Filtered Robust Tracking Control of Wave-Adaptive Modular Vessel with Uncertainty,” *Proceedings of the 2020 American Control Conference (ACC)*, IEEE, Denver, CO, 2020, pp. 2118–2123. <https://doi.org/10.23919/ACC45564.2020.9147520>
- [26] Dony, M. N.-A.-A., and Arhin, B., “Safe Model-Free Q-Learning for Discrete-Time Fully Cooperative Multi-Input Systems with State and Control Constraints via Control Barrier Functions,” Research Square preprint, 2025, submitted for publication. <https://doi.org/10.21203/rs.3.rs-9108014/v1>
- [27] Dony, M. N.-A.-A., Khallil, M. E., and Ali, M. S., “Adaptive Reinforcement Learning with Control Barrier Functions for Safe State Avoidance in Discrete Event Systems,” *Proceedings of the 8th IEOM Bangladesh International Conference on Industrial Engineering and Operations Management*, Dhaka, Bangladesh, 2025. <https://doi.org/10.46254/BA08.20250300>
- [28] Stevens, B. L., Lewis, F. L., and Johnson, E. N., *Aircraft Control and Simulation: Dynamics, Controls Design, and Autonomous Systems*, 3rd ed., Wiley, Hoboken, NJ, 2016.
- [29] Etkin, B., and Reid, L. D., *Dynamics of Flight: Stability and Control*, 3rd ed., Wiley, New York, 1995.
- [30] Beard, R. W., and McLain, T. W., *Small Unmanned Aircraft: Theory and Practice*, Princeton University Press, Princeton, NJ, 2012.
- [31] Anderson, J. D., *Aircraft Performance and Design*, McGraw-Hill, New York, 1999.
- [32] Selig, M. S., Guglielmo, J. J., Broeren, A. P., and Giguère, P., *Summary of Low-Speed Airfoil Data, Volume 1*, SoarTech Publications, Virginia Beach, VA, 1995.
- [33] Maughmer, M. D., Axten, R. M., and Metkowski, L. P., “A Wind Tunnel for the Study of Low-Reynolds-Number Airfoils,” *Journal of Aircraft*, Vol. 60, No. 6, 2023, pp. 1739–1745.
- [34] Hann, R., Hearst, R. J., Sætran, L. R., and Bracchi, T., “Experimental and Numerical Icing Penalties of an S826 Airfoil at Low Reynolds Numbers,” *Aerospace*, Vol. 7, No. 4, 2020, Paper 46. <https://doi.org/10.3390/aerospace7040046>
- [35] Menter, F. R., “Two-Equation Eddy-Viscosity Turbulence Models for Engineering Applications,” *AIAA Journal*, Vol. 32, No. 8, 1994, pp. 1598–1605. <https://doi.org/10.2514/3.12149>

- [36] Dony, M. N.-A.-A., and Arhin, B., “Data-Driven Distributed Battery State-of-Charge Balancing Using Neural Network-Based Open Circuit Voltage Learning,” *IFAC Journal of Systems and Control*, 2026. <https://doi.org/10.1016/j.ifacsc.2026.100415>
- [37] Khallil, M. E., Islam, M. A., Dony, M. N.-A.-A., and Ali, M. S., “Comparative Analysis of LSTM and PINN for Internal Resistance Estimation in Lithium-Ion Batteries,” *Proceedings of the 8th IEOM Bangladesh International Conference on Industrial Engineering and Operations Management*, Dhaka, Bangladesh, 2025. <https://doi.org/10.46254/BA08.20250259>